In [1]:
```
import numpy as np
```

In [2]:
```
X = np.array([[1,2],[2,4],[3,6]])
y = np.array([1,2,3])

lam = 0.5
```

In [3]:
```
x_mean = np.mean(X, axis=0)
y_mean = np.mean(y)
```

In [4]:
```
X = X-x_mean
y = y-y_mean
```

In [5]:
```
X
```

Out[5]:
```
array([[-1., -2.],
       [ 0.,  0.],
       [ 1.,  2.]])
```

In [6]:
```
y
```

Out[6]:
```
array([-1.,  0.,  1.])
```

In [7]:
```
I = np.eye(X.shape[1])
I
```

Out[7]:
```
array([[1., 0.],
       [0., 1.]])
```

In [8]:
```
beta_cap = np.dot(np.linalg.inv(np.dot(X.T,X) + lam * I), np.dot(X.T,y))
beta_cap
```

Out[8]:
```
array([0.19047619, 0.38095238])
```

In [9]:
```
beta_0_cap = y_mean - np.dot(x_mean, beta_cap)
beta_0_cap
```

Out[9]:
```
0.0952380952380949
```

In [10]:
```
y_pred = beta_0_cap + np.dot(X, beta_cap)
y_pred
```

Out[10]:
```
array([-0.85714286,  0.0952381 ,  1.04761905])
```

In [11]:
```
RSS = np.sum((y - y_pred) ** 2)
```

In [12]:
```
print(RSS)
```

```
0.03174603174603152
```

In [ ]: